# Siddharth R

B.Tech Computer Science and Engineering
SRM Institute of Science and Technology

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## Introduction

Building AI and machine learning systems through research and experimentation, with projects spanning reinforcement learning, multi-agent systems, and anomaly detection. Interested in accelerating AI research and research adoption.

## Certifications & Coursework

Machine Learning Specialization, Deep Learning Specialization, Probability and Statistics for ML, Stanford CS229, Stanford CS231N

## Technical Skills

- **Languages:** Python, Java, JavaScript, C
- **Machine Learning:** PyTorch, Scikit-learn, Deep Learning, Reinforcement Learning
- **Frameworks & Tools:** React, Node.js, Express.js, MongoDB, Git
- **Areas of Interest:** World Models, Intelligent Agents, ML Systems

## Projects

### DreamerV1 World Model

*Latent imagination-based reinforcement learning on Gridworld* — PyTorch, LightningAI

- Implemented DreamerV1 from scratch to study world model behavior on discrete Gridworld tasks, isolating failure modes not evident in Atari benchmarks.
- **Authored a research article** analyzing these failure modes; findings provide actionable guidance for adapting world models to non-continuous state spaces.

### Paper2Brain

*Research paper to interactive knowledge graph system* — Agno, Groq, LlamaParse, React Flow

- Architected a **multi-agent LLM pipeline** using Agno and Groq to extract semantically structured knowledge graphs from academic papers, reducing dense paper comprehension time.
- Implemented RAG-based querying with critique-feedback loops for iterative semantic correction, improving graph accuracy over baseline single-pass extraction.

### SafarX

*Tourist behavior modeling and anomaly detection system* — NumPy, Scikit-learn, React

- Designed a **mathematical generative model** to synthesize realistic tourist behavior data, enabling robust training without real-world data collection overhead.
- Achieved **>95% accuracy** on detecting anomalous behaviors (prolonged inactivity, network drop-off) using statistical techniques; minimized false positives for real-world safety deployment.

### Atari with DQN

*Faithful reimplementation of DeepMind's seminal DQN paper* — PyTorch, Google Colab

- Reproduced "Playing Atari with Deep Reinforcement Learning" end-to-end, including experience replay and target networks, as a deep-dive into foundational RL methodology.
- Trained on Breakout using a T4 GPU via Colab; agent achieved stable, **recognizable gameplay within 6 hours** of wall-clock training time.

## Hackathons & Competitions

### HeisenHack 2025 — Top 10 / 700+ Teams

*Tourist Safety AI Pipeline*

- Built an end-to-end anomaly detection system for tourist safety, integrating behavior modeling with a real-time threat-alerting pipeline.
- Outperformed 700+ competing teams; system was recognized for accuracy and low-latency alert delivery.

### NIT-Trichy Pragyan26 — Top 10 / 200+ Teams

*AI Triage Management Software*

- Designed an AI-powered patient prioritization framework for high-load hospital scenarios.
- Finished in the top 10 of 200+ teams.

## Publications

- **LLM-Assisted Autonomous Research for Solver-Guided VLNS Search in Connected Still Life** *Manuscript in preparation*

## Education

**SRM Institute of Science and Technology** — *CGPA: 9.1*

*B.Tech Computer Science and Engineering* — Expected 2028